

When you send a letter, you drop it off at your local post office. Your local post office looks at the address and figures out which regional post office the letter should go to. Then the regional post office looks at the address and finds out which local post office is closest to your recipient. Finally, the recipient's local post office delivers your letter to its recipient.

Computers are like post offices, and the simple mail transport protocol (SMTP) is the procedure, rather a protocol which a post office uses to find out where to send the letter next in the process. Any program that sends an e-mail message uses SMTP protocol (default port 25) to deliver that message to the next 'post office' for relaying to its final destination.

Now, we have sent the mail using SMTP. What about retrieving the e-mail or reading the e-mail? Well for this, two common protocols are used, which are namely POP3 (post office protocol) (default port 110) and IMAP3 (Internet message access protocol) (default port 143). The working of e-mail is depicted in Fig. 1.

In today's scenario, the objective for the user might be summarised as follows (borrowing from the paper world): To be certain what they send goes to the right person/place; to be certain that the right person/place can read the information; to be able to use signed receipt as proof to a court or other body; to stop the wrong people from reading personal and private information. Some of these wishes are more difficult than others.

Just as in the paper world you cannot stop anyone from seeing the address on the letters' envelope, the same is true for an e-mail. If someone alters that address, it does not go to the right place, and if someone alters the return address, the recipient may not know where it has come from, or it may not, if delivery fails, be returned to the correct sender.

We are familiar with the paper world and it has some benefits. You can usually see if someone has already opened your mail. The post office can

often cope with wrong addressing and still get it to the right place. E-mail is rather different. There is no way of telling who reads the mail unless you take actual steps to make it impossible. The e-mail post office cannot cope with any address errors whatsoever. It has no idea if any of the addresses on the mail are correct and cannot tell if they have been altered. There is no plain envelope to stop people reading the contents and it is possible for

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When we use some tool on our personal computer for an Internet connection to communicate with a Web server, the language that the Internet connection uses is HTTP—hypertext transfer protocol. When we send our message with Web mail, the Web server contacts its SMTP server and sends our message to it.

To summarise, when we use the Web-based e-mail programs like Gmail (which is also a cloud service), the Web server already has the SMTP/POP3/IMAP3 address inserted. If we want to access other e-mail accounts maintained by any other e-mail server (term used for SMTP/POP3/IMAP3 server), it can be done using these Web-based services, by mentioning the specific

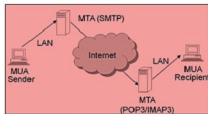


Fig. 1: E-mail flow diagram

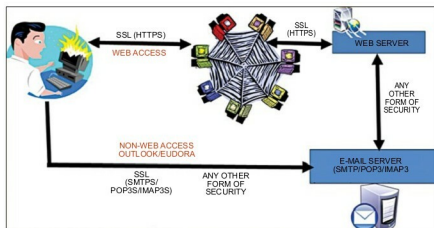


Fig. 2: Secure e-mail access with and without Web server

hackers, government agencies and almost anyone else to read the mail. Proof of delivery is worth the paper it is printed on.

Most people send mail in two ways—with a Web-based interface like Yahoo!, Gmail or Hotmail, or with an e-mail client program like Outlook or Eudora. When we send a message with an e-mail program on our personal computer (or cell phone or PDA), we have to specify a server so that our e-mail program knows where to send the message. This server is like our local post office. An e-mail program talks directly to the server using the computer protocol (language) known as SMTP.

e-mail server address. Let us see the security in both these cases.

In case of e-mail clients like Outlook or any other non-Web-based application, we can apply security on the message and send the e-mail. This kind of security is shown in Fig. 2. In case of Web-based scenario, we can have an SSL kind of environment between client and Web server, and SSL or some other kind of security between the Web server and the e-mail server. Between Web server and client, we can invoke SSL in the form of <https://www.site.com>. Here SSL, which runs at the transport layer (default port 443), takes care of all the communication be-